mgcvUI GAM Report

2026-03-25

Model Overview

Metric	Value
R-squared	0.8331
Deviance Explained	83.7%
AIC	4.31261×10^4
BIC	4.3341×10^4
Observations	1488
Smooth Terms	6
Method	REML
Family	gaussian
Fitting Time	0.75 sec

Formula

```
sale_price ~ s(sale_age, bs = "cr", k = 6) + s(living_sqft, bs = "tp") + s(beds_total,      bs = "tp", k
```

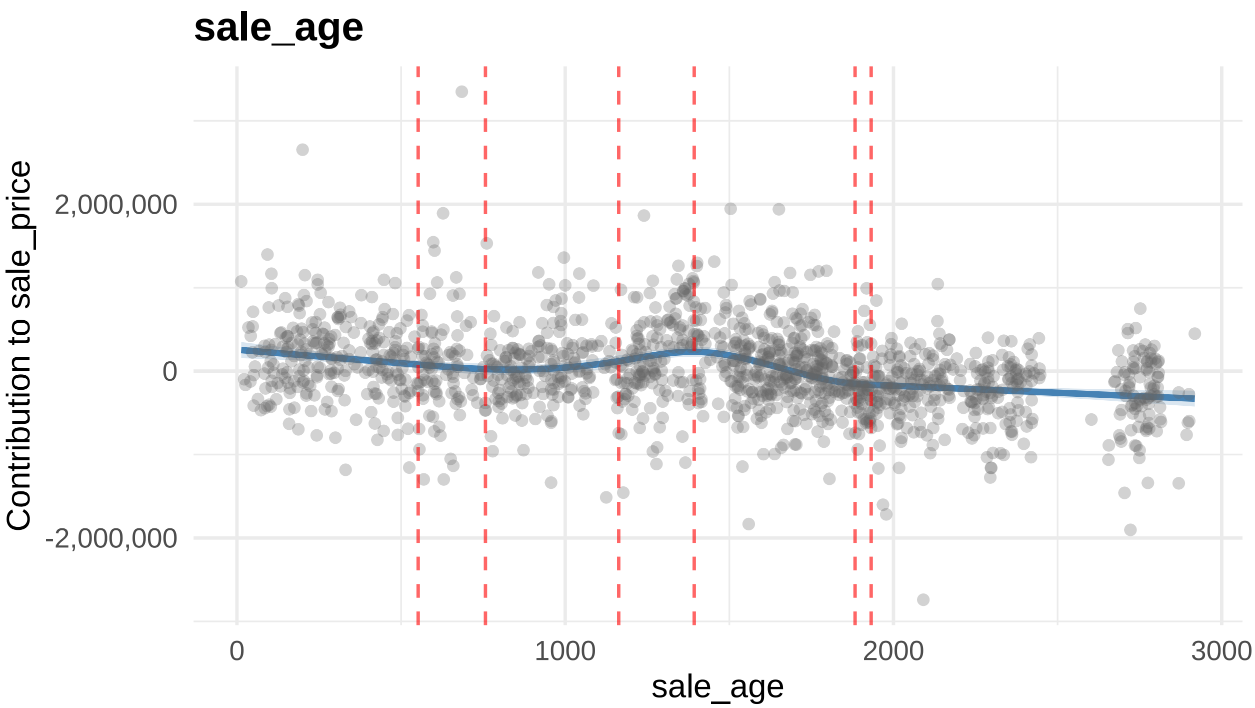
Summary

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## sale_price ~ s(sale_age, bs = "cr", k = 6) + s(living_sqft, bs = "tp") +
##     s(beds_total, bs = "tp", k = 8) + s(baths_total, bs = "cr",
##     k = 4) + lot_size + area_id + s(age, bs = "tp") + latitude +
##     longitude + s(garage_spaces, bs = "tp", k = 5)
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -4.051e+08  1.131e+08  -3.582 0.000353 ***
## lot_size     4.635e+00  2.901e+00   1.598 0.110344
## area_id     -2.202e+04  5.700e+03  -3.863 0.000117 ***
## latitude    -2.945e+06  2.170e+06  -1.357 0.174931
## longitude   -4.356e+06  1.201e+06  -3.627 0.000297 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                    edf Ref.df       F  p-value
## s(sale_age)      4.564      5  38.196  < 2e-16 ***
## s(living_sqft)   8.598      9 118.145  < 2e-16 ***
## s(beds_total)    6.881      7  59.634  < 2e-16 ***
```

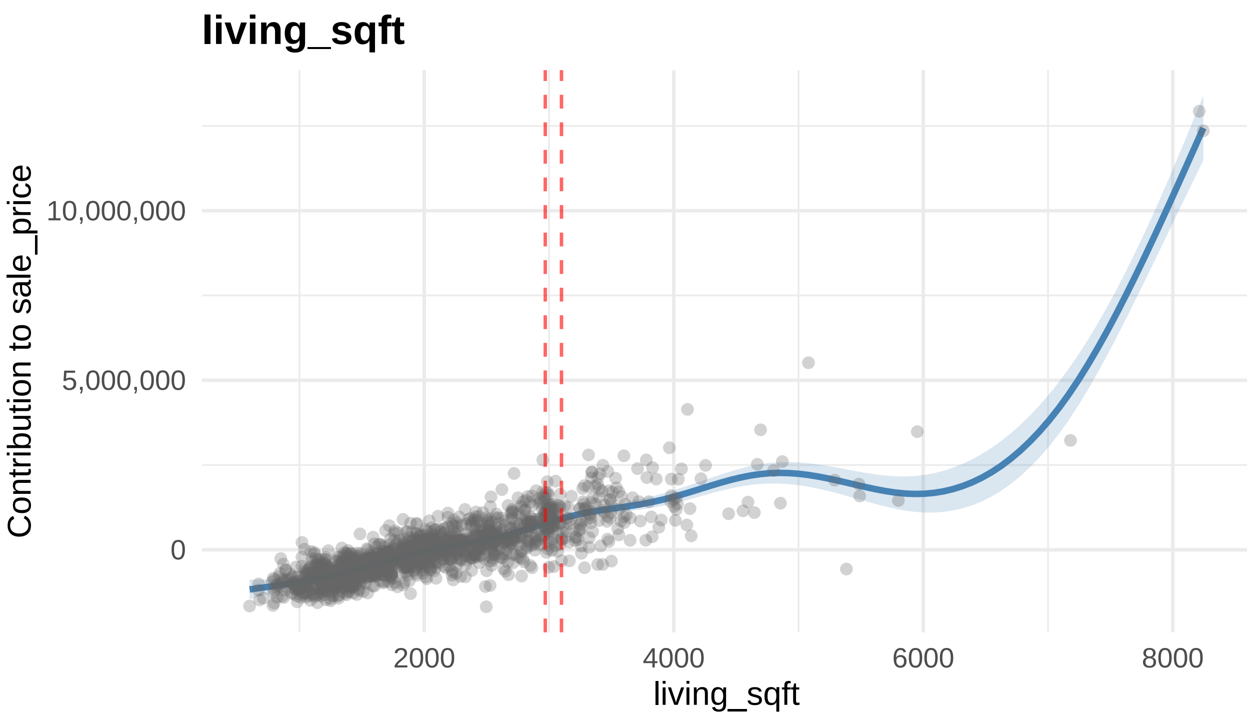
```
## s(baths_total)   2.525      3  13.891  < 2e-16 ***
## s(age)           6.841      9  27.194  < 2e-16 ***
## s(garage_spaces) 2.475      4   3.938 0.000322 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.833   Deviance explained = 83.7%
## -REML =  15415  Scale est. = 2.1967e+11  n = 1488
```

Smooth Term Plots

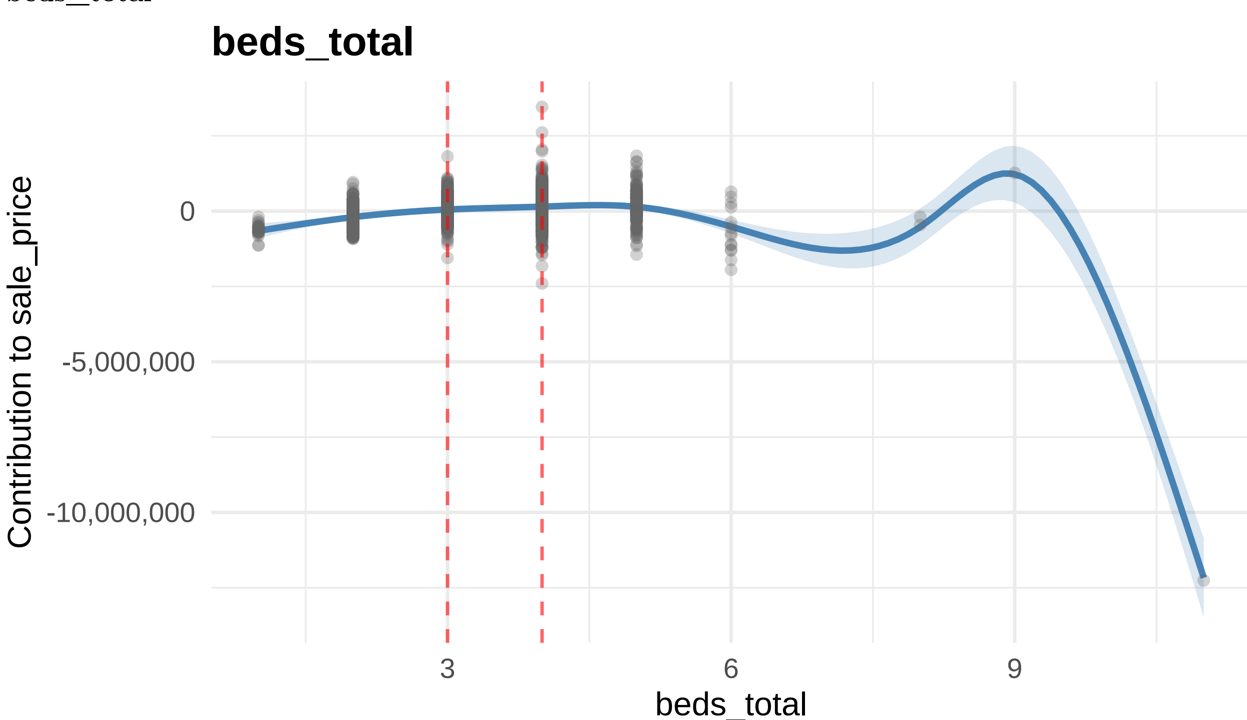
sale_age



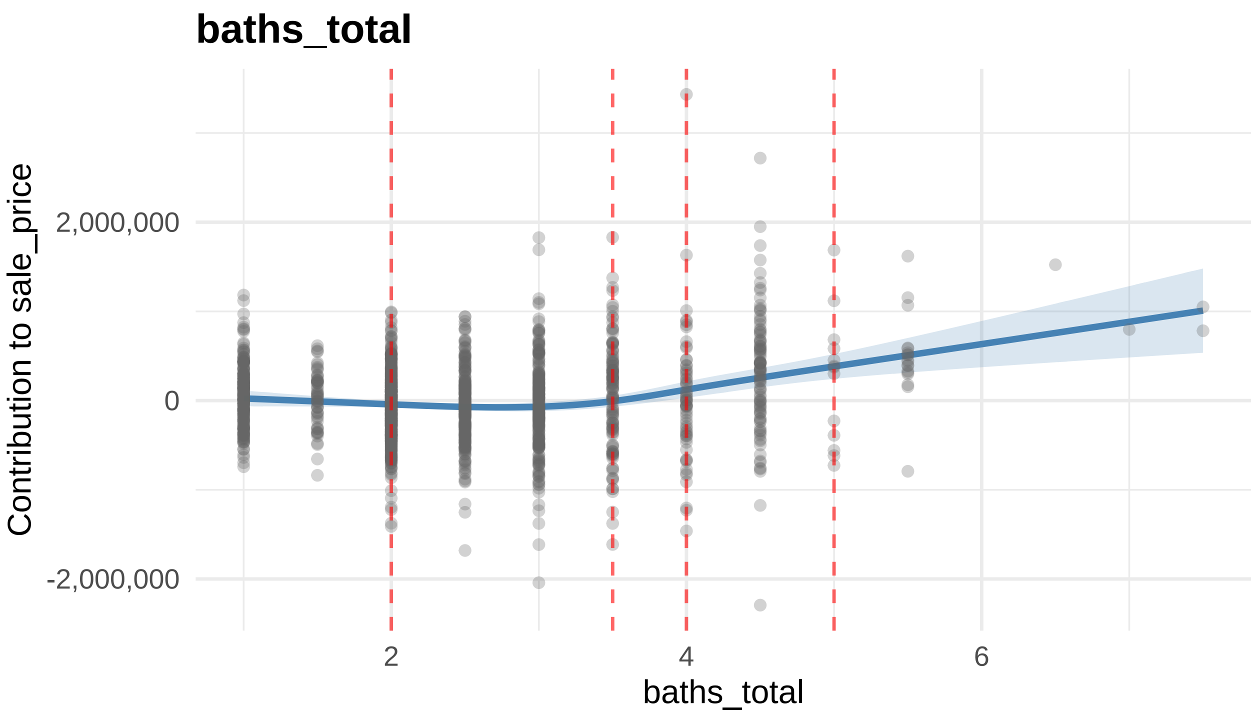
living_sqft



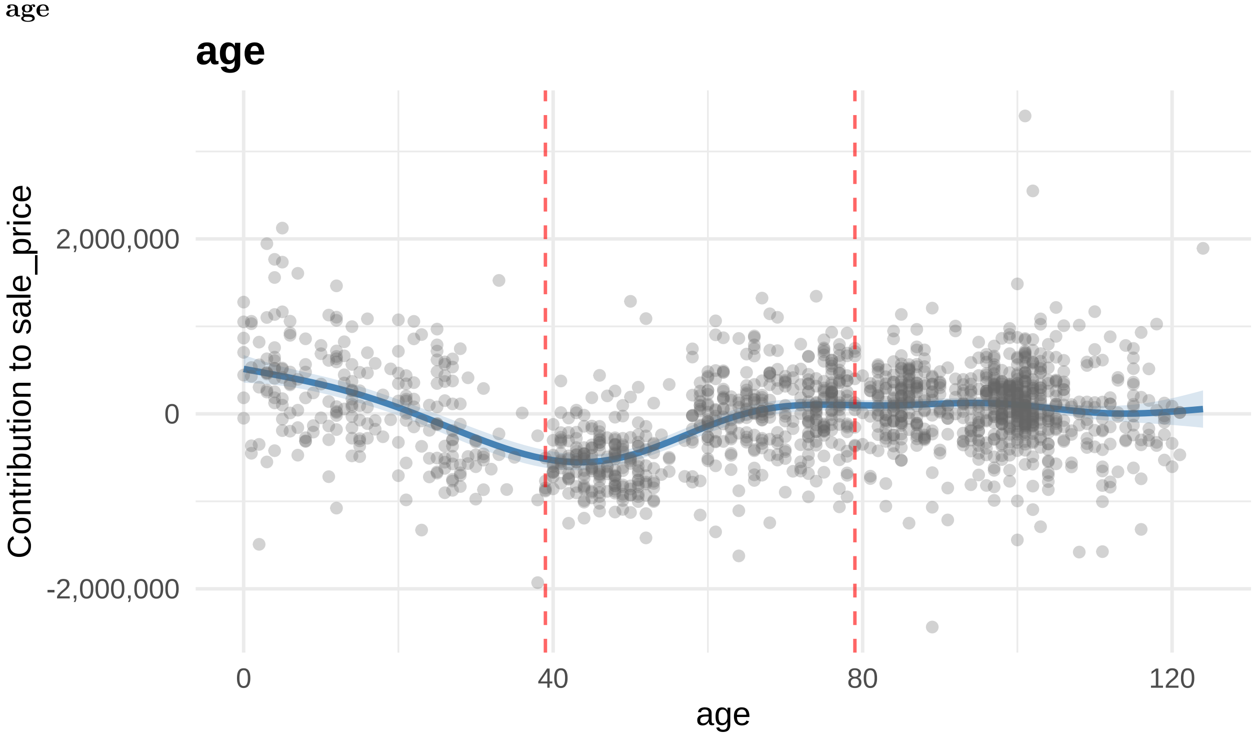
beds_total



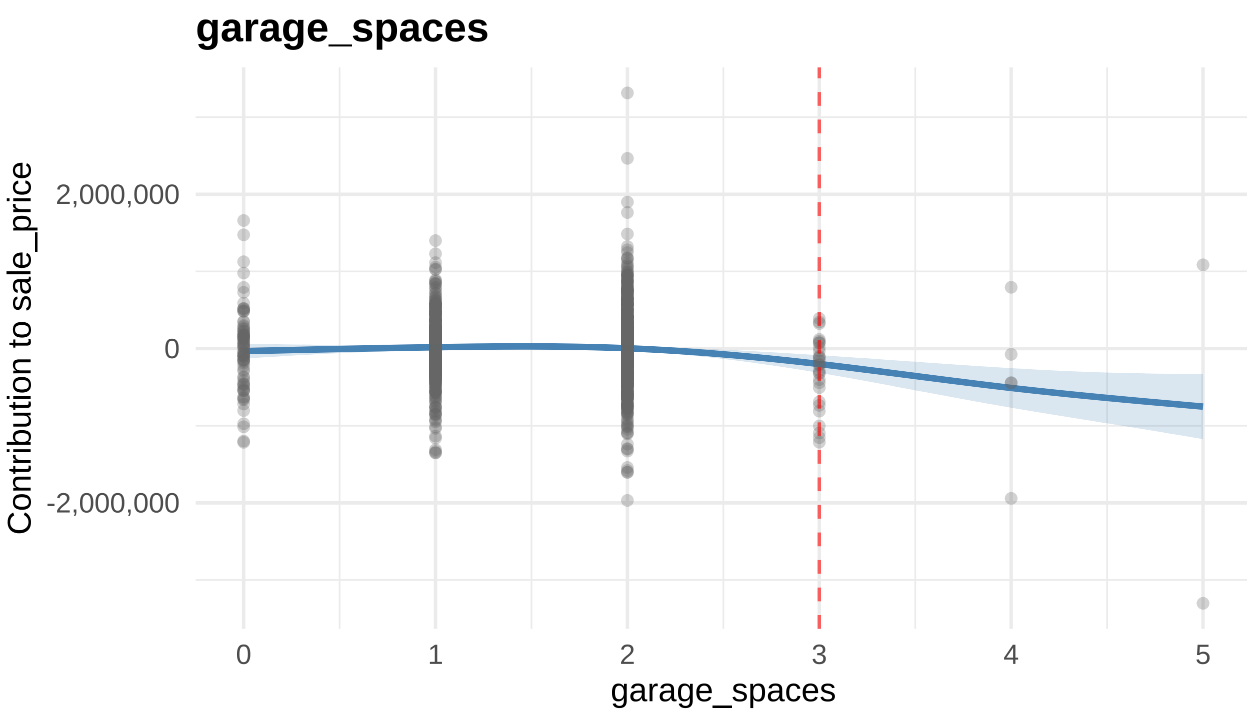
baths_total



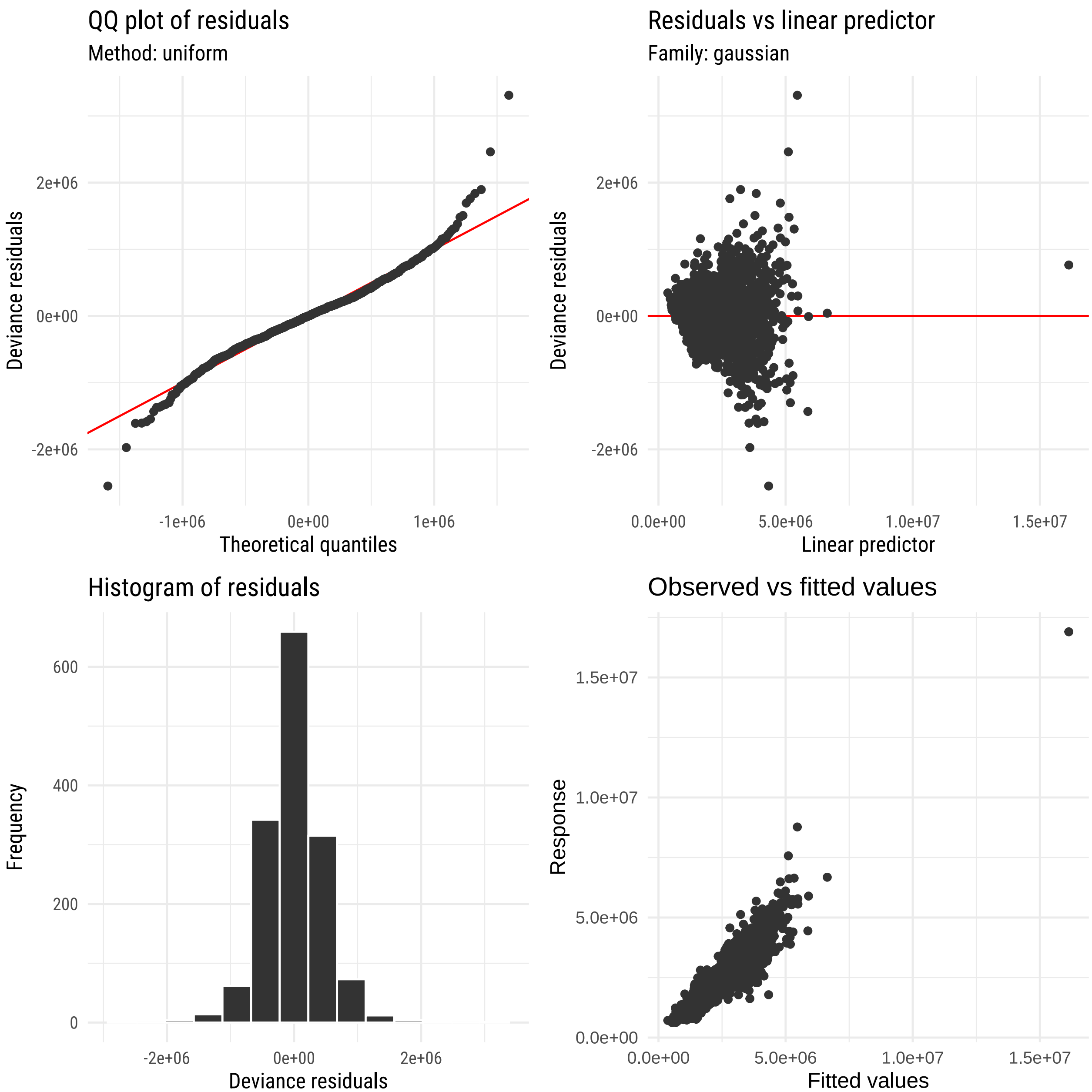
age



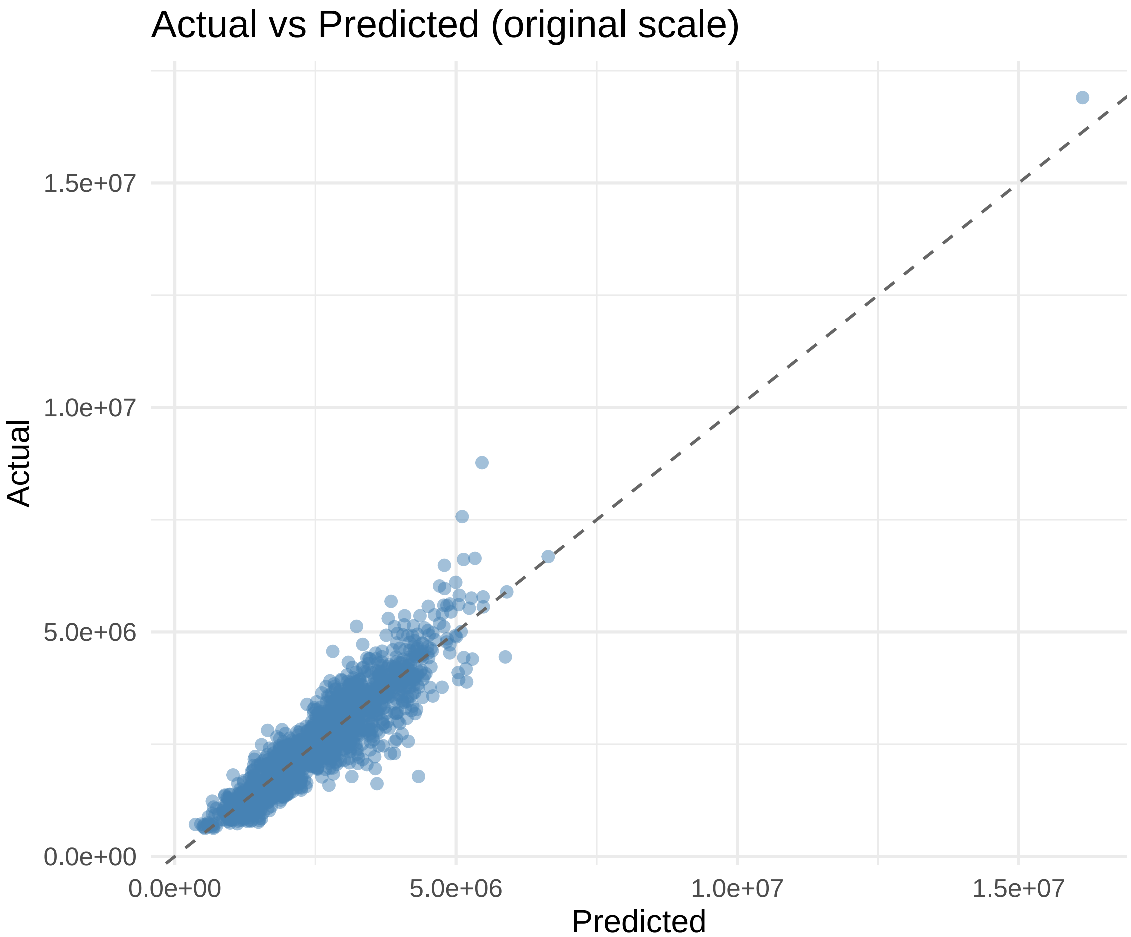
garage_spaces



Diagnostics



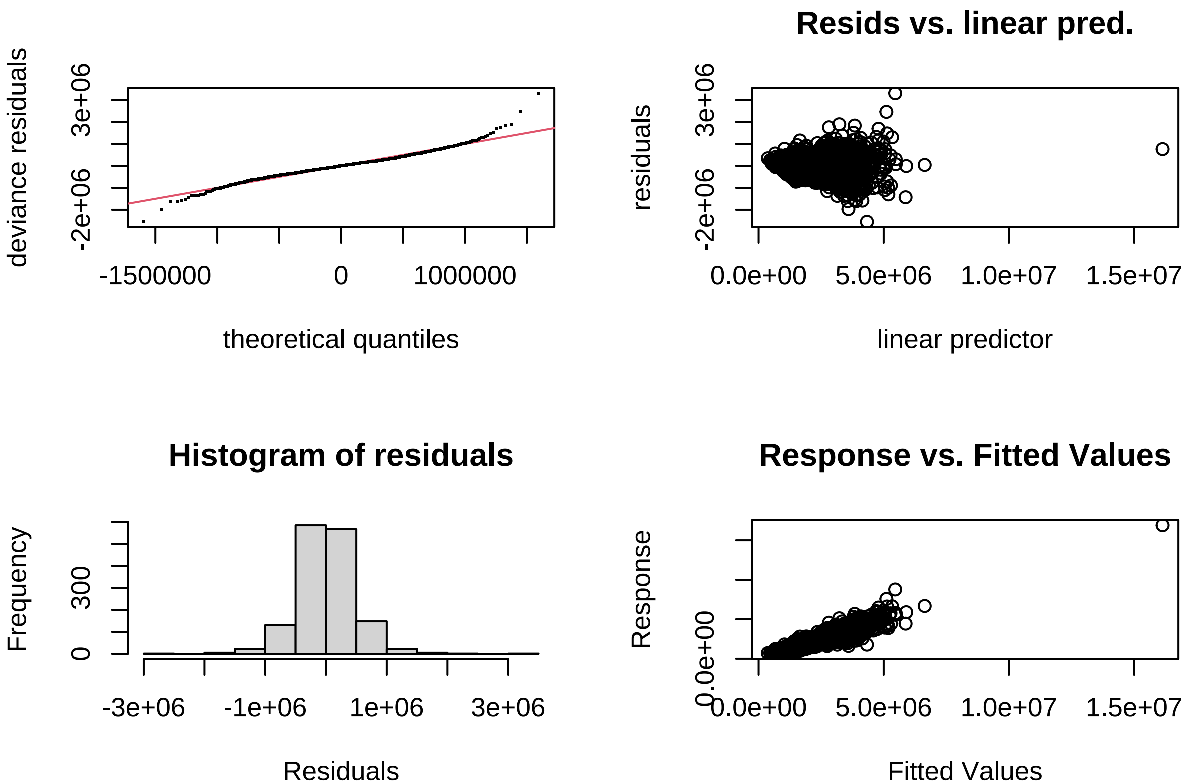
Actual vs Predicted



Earth vs GAM Sign Consistency

variable	earth_direction	gam_direction	consistent
sale_age	increasing	decreasing	FALSE
living_sqft	increasing	increasing	TRUE
beds_total	mixed	mixed	TRUE
baths_total	increasing	increasing	TRUE
age	increasing	mixed	TRUE
latitude	increasing	mixed	TRUE
longitude	mixed	mixed	TRUE
garage_spaces	increasing	decreasing	FALSE

Basis Dimension Check




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 18 iterations.
## Gradient range [-0.001872508,0.001820109]
## (score 15415 & scale 219674838414).
## Hessian positive definite, eigenvalue range [0.06006226,529.0055].
## Model rank =  42 / 42
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                    k'  edf k-index p-value
## s(sale_age)      5.00 4.56    0.97   0.145
## s(living_sqft)   9.00 8.60    1.00   0.425
## s(beds_total)    7.00 6.88    0.86  <2e-16 ***
## s(baths_total)   3.00 2.53    0.92  <2e-16 ***
## s(age)           9.00 6.84    0.90  <2e-16 ***
## s(garage_spaces) 4.00 2.48    0.96   0.075 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```